

Quantitative Methods for Business

Extra Assignment

Daniel Sánchez Pazmiño

July 15, 2026

1. Write down the gradient and the Hessian for each of the following functions:

(a) $f(x, y) = x^4 + x^2 - 6xy + 3y^2$

(b) $f(x, y) = xy^2 + x^3y - xy$

(c) $f(x, y, z) = x^2 + 6xy + y^2 - 3yz + 4z^2 - 10x - 5y - 21z$

Note: The Hessian collects the second partial derivatives, as follows:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{bmatrix}$$

2. Consider the function $f(x) = -\frac{x^3}{3} + x^2 - x + 4$.
 - (a) Draw the graph of $f(x)$.
 - (b) Find the critical points of $f(x)$.
 - (c) Determine the concavity of $f(x)$.
 - (d) Find any local maxima or minima of $f(x)$, if they exist.
 - (e) If I were to restrict the domain of $f(x)$ to $[3, 3]$ what would be the maximum and minimum values of $f(x)$? Are these values local maxima or minima?
3. Consider demand for hospital beds in a city, given by $Q = 100 - 2P$, where Q is the quantity of hospital beds demanded and P is the price of a hospital bed.
 - (a) When COVID-19 hit the city, the demand for hospital beds increased and shifted right to $Q = 200 - 2P$. Draw the graph of the demand curve before and after COVID-19.

- (b) If the supply of hospital beds is given by $Q = 50 + 3P$, find the equilibrium price and quantity of hospital beds before and after COVID-19.
 - (c) If the city government wants to increase the quantity of hospital beds to 300, what should the price be?
 - (d) A reduced form estimation of demand for hospital beds in the city is given by $Q = 100 - 2P + 0.5I$, where I is the income of the city. Calculate the partial price elasticity of demand for hospital beds in the city at equilibrium, with an Income I of 100. Interpret the result.
 - (e) Assume the government intervenes and increases the price by 20%. Calculate the percentage change in quantity demanded of hospital beds. Interpret the result.
4. Consider the sum of squared errors, a statistical measure that will be covered later in the program. It is expressed as

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual value of the dependent variable, $\hat{y}_i$ is the predicted value of the dependent variable, and n is the number of observations.

- (a) Using properties of sums, show that the sum of squared errors can be expressed as

$$SSE = \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n y_i \hat{y}_i + \sum_{i=1}^n \hat{y}_i^2$$